Progress Note: 2-Period Igami Model with Agglomeration

14.273 Mini Project

April 25, 2026

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1 Overview

This note documents the first extension of Igami (2017) to incorporate **innovation spillovers / agglomeration effects** in a simplified 2-period setting. The goal is to understand whether these effects undo the innovator's dilemma found by Igami.

2 Model

2.1 State and Firm Types

The market state at each period t is $s_t = (n_o, n_b, n_n, n_{pe})$, where:

- n_o: old incumbents (have *not* adopted new technology),
- n_b: "both" incumbents (have adopted; produce with new tech),
- n_n: new entrants (using new technology),
- n_{pe}: potential entrants.

Active firms in the product market are those with $n_o + n_b + n_n > 0$.

"Both" firms are multi-product: they continue to sell old-gen (at cost c_o) while also selling new-gen (at cost $c_{n,t}$). This is the source of the **cannibalization effect**: by expanding new-gen supply, a "both" firm depresses the old-gen price it receives, creating a disincentive to innovate that is absent for pure new entrants.

2.2 Timing Within Each Period

1. **Product market:** Cournot competition in *two differentiated markets* (old-generation and new-generation products) with linear inverse demand:

$$P_o = A - \frac{B}{M} Q_o - \frac{\rho B}{M} Q_n, \tag{1}$$

$$P_n = A - \frac{B}{M} Q_n - \frac{\rho B}{M} Q_o, \tag{2}$$

where $\rho \in [0, 1)$ is the cross-market substitution parameter. Old firms (n_o) compete only in the old-gen market at cost c_o. New entrants (n_n) compete only in the new-gen market at cost $c_{n,t}$. "Both" firms (n_b) are *multi-product*: they sell in both markets simultaneously and internalize the **cannibalization** effect, in that their new-gen output reduces demand for their own old-gen product.

2. **Private shocks:** Each firm draws i.i.d. cost shocks $\varepsilon_{i,a} \sim \mathrm{EVT1}(0, \sigma)$ for each action a.

3. **Discrete choices** (sequential order: old $\to$ both $\to$ new $\to$ pe):

   - old: $\{\text{stay}, \text{innovate}, \text{exit}\}$
   - both/new: $\{\text{stay}, \text{exit}\}$
   - pe: $\{\text{enter}, \text{quit}\}$

   Each type observes the *realized* outcomes of all earlier movers before making its own decision.

4. **Transition** to period $t + 1$.

2.3 Agglomeration Extension

The new-technology marginal cost depends on the number of firms already using it:

$$c_{n,t} = c_{n,0} - \gamma \cdot (n_{b,t} + n_{n,t}), \qquad \gamma \geq 0. \tag{3}$$

When $\gamma = 0$ the model reduces to the Igami baseline. When $\gamma > 0$, a larger innovative cluster lowers costs for all new-tech firms, capturing the knowledge-spillover or agglomeration channel.

2.4 Value Functions and CCPs

Because firms move sequentially within each period and each type observes the *realized* outcomes of all earlier movers, each type's decision problem is effectively a *single-agent problem*: earlier-mover uncertainty has been resolved, and later-mover choices enter only through pre-computed CCPs. Combined with i.i.d. EVT1 private cost shocks, this yields closed-form logit CCPs at each stage:

$$P(a \mid \tau, \tilde{s}) = \frac{\exp\bigl(\bar{V}_a(\tau, \tilde{s})/\sigma\bigr)}{\sum_{a'} \exp\bigl(\bar{V}_{a'}(\tau, \tilde{s})/\sigma\bigr)}, \tag{4}$$

where $\tilde{s}$ is the *realized intermediate state* after all earlier movers have acted, and $\bar{V}_a$ is the systematic utility:

$$\bar{V}_a(\tau, \tilde{s}) = \pi_\tau(s_0) + \mathrm{cost}(a) + \beta\, \mathbb{E}[V_{t+1}(\tau', s') \mid \tilde{s},\, a_i = a,\, \mathrm{CCP}_{\mathrm{later}}]\,. \tag{5}$$

Here $\mathrm{CCP}_{\mathrm{later}}$ denotes pre-computed CCPs of same-type peers and all later-moving types; flow profits $\pi_\tau(s_0)$ are evaluated at the period-opening state.

2.5 Solution Method

The model is solved by backward induction, following Igami (2017):

1. **Terminal period ($t = 1$):** $V_1(\tau, s_1) = \pi_\tau(s_1)$, where Cournot profits use the agglomeration-adjusted cost $c_{n,1} = c_{n,0} - \gamma(n_{b,1} + n_{n,1})$.

2. **Period $t = 0$:** For each state s_0, stages are solved in *reverse move order* (PE $\rightarrow$ new $\rightarrow$ both $\rightarrow$ old), exploiting the sequential structure:

   - **Stage 4 (PE):** Each PE firm observes the full realized intermediate state $\tilde{s}$ and solves a binary logit $\{\text{enter}, \text{quit}\}$. We compute CCPs for all intermediate states reachable from s_0.
   - **Stage 3 (new):** Each new firm solves a binary logit $\{\text{stay}, \text{exit}\}$, conditioning on pre-computed PE CCPs to integrate over the PE-stage continuation value.
   - **Stage 2 (both):** Each both firm solves a binary logit, conditioning on new + PE CCPs. Innovators locked in from Stage 1 (k_{io} old firms that chose innovate) are added to n_b before this stage.
   - **Stage 1 (old):** Each old firm solves a multinomial logit $\{\text{stay}, \text{innovate}, \text{exit}\}$, conditioning on all lower-stage CCPs.

   No fixed-point iteration is needed within the period: the sequential structure makes each stage a closed-form single-agent problem given pre-computed downstream CCPs. Marginal CCPs for later-moving types are recovered by integrating over the equilibrium transition distribution (products of independent Binomial / Multinomial draws).

3 Calibration

We summarize the parameters we use for calibration and for simulating the data in Table 1.

The initial state of interest in the non-spatial baseline is $s_0 = (n_o, n_b, n_n, n_{pe}) = (4, 1, 1, 2)$, loosely inspired by the early HDD market structure in Igami (2017).

4 Results

4.1 Baseline vs. Agglomeration

Table 2 compares equilibrium CCPs at s_0 for $\gamma = 0$ (baseline) and $\gamma = 0.05$.

Agglomeration raises both the innovation and entry rates. Intuitively, a larger innovative cluster lowers future costs for new-tech firms, raising the expected continuation value of innovating or entering.

Table 1: Parameter values for calibration and simulation

Parameter	Description	Value
A	Demand intercept	3.0
B	Demand slope	1.0
M	Market size	1.0
c_o	Old-tech marginal cost	1.5
$c_{n,0}$	New-tech baseline cost	0.5
β	Discount factor	0.9
κ	Innovation cost	0.3
ϕ	Entry cost	0.2
σ	EVT1 scale	0.5
γ	Agglomeration (default)	0.15
ρ	Cross-market substitution	0.5

Table 2: Baseline vs. agglomeration

	$\gamma = 0$	$\gamma = 0.05$
$P(\text{innovate} \mid \text{old})$	0.3135	0.3206
$P(\text{enter} \mid \text{pe})$	0.5061	0.5139

Notes: Equilibrium CCPs at $s_0 = (4, 1, 1, 2)$ for the baseline ($\gamma = 0$) and agglomeration ($\gamma = 0.05$) specifications.

4.2 Comparative Statics in γ

Table 3 reports equilibrium CCPs as γ varies from 0 to 0.30, and Figure 1 plots the same.

4.3 Comparative Statics in ρ (Cannibalization)

Figure 2 plots equilibrium CCPs as the substitution parameter ρ varies from 0 (independent markets) to 0.95 (near-homogeneous goods), holding $\gamma = 0.05$.

4.4 Discussion

Three features of the comparative statics are notable:

1. **Positive agglomeration effect:** Higher γ unambiguously raises the innovation rate for small values of γ. This confirms the model's central prediction: knowledge spillovers encourage adoption.

2. **Cannibalization deters innovation:** Higher ρ (stronger substitution between old-gen and new-gen) reduces the profit of "both" firms by depressing old-gen prices. In

Table 3: Comparative statics in γ

γ	P(innovate|old)	P(enter|pe)
0.00	0.3135	0.5061
0.05	0.3206	0.5139
0.10	0.3282	0.5221
0.15	0.3314	0.5258
0.20	0.3318	0.5263
0.25	0.3315	0.5260
0.30	0.3316	0.5260

Notes: Equilibrium CCPs at $s_0 = (4, 1, 1, 2)$ as the agglomeration parameter γ varies from 0 to 0.30.

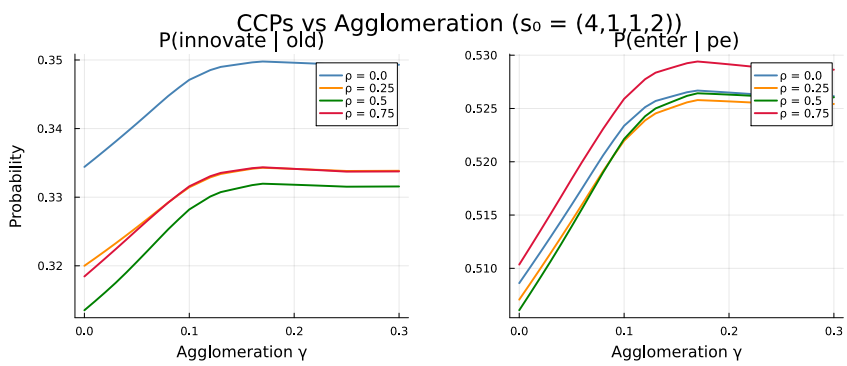

Figure 1: P(innovate | old) and P(enter | pe) as functions of γ at $s_0 = (4, 1, 1, 2)$.

the calibration, per-firm "both" profit falls from ≈ 0.531 at $\rho = 0$ to ≈ 0.327 at $\rho = 0.5$. This is the central mechanism in Igami (2017): incumbents are reluctant to innovate because doing so cannibalizes their existing old-gen revenue. The relation is nonmonotone, however: as ρ approaches 1, the two markets become nearly homogeneous and the incumbent benefits from innovating because it can sell more of the new-gen product.

3. **Saturation:** The innovation rate saturates near 1/3 for large γ. This is a feature of the EVT1 / logit structure: with large σ relative to payoff differences, all actions approach equal probability. In a richer calibration (smaller σ), the effect would be more pronounced.

5 Regional Extension

5.1 Regional Structure

To study the interaction between industrial policy and agglomeration we extend the state space with a **region** attribute. There are $R = 3$ regions (labelled $r \in \{1, 2, 3\}$); each firm is born into a region and never leaves it, and potential entrants sit in per-region pools $n_{pe,r}$. Product-market competition stays **global**, with all active firms meeting in the same two-

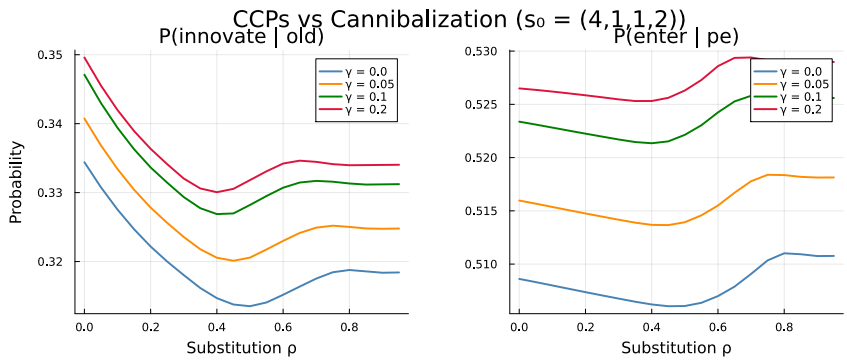

Figure 2: $P(\text{innovate} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ as functions of ρ at $s_0 = (4, 1, 1, 2)$, $\gamma = 0.05$.

market Cournot game of equations (1)–(2). Agglomeration spillovers, in contrast, are now **local**:

$$c_{n,r,t} = c_{n,0} - \gamma_r \cdot \big(n_{b,r,t} + n_{n,r,t}\big), \qquad r = 1, \ldots, R. \tag{6}$$

Because $c_{n,r,t}$ varies across regions, firms of type "both" and "new" face *heterogeneous* marginal costs in the new-gen market. The Cournot stage therefore loses symmetry across regions, and equilibrium quantities are obtained by solving an asymmetric FOC system of up to $4R$ equations per state. Within-region slots remain symmetric (all firms in the same $(type, r)$ cell pick the same quantity), so per-firm profits are still characterised by $\pi_{o,r} = bq^2_{oo,r}$, $\pi_{n,r} = bq^2_{nn,r}$, and $\pi_{b,r} = b(q^2_{bo,r} + q^2_{bn,r}) + 2s\, q_{bo,r} q_{bn,r}$ where $b = B/M$ and $s = \rho B/M$.

Each firm privately captures the lower cost when it innovates but does not internalize the benefit it confers on regional peers. This is a standard **local agglomeration externality**, creating a wedge between the social and private return to innovation and opening the door to regionally targeted industrial policy.

5.2 Solution with Regional States

The sequential-move structure (old $\to$ both $\to$ new $\to$ pe) carries over unchanged, and we extend the same idea one level down: **within each stage, regions move sequentially**, $r = 1 \to 2 \to 3$. When firms in region r decide, they observe the realized outcome of regions $1, \ldots, r-1$ and integrate over the (yet-to-play) strategies of regions $r+1, \ldots, R$. This is the regional analogue of Igami's cross-type sequencing: it makes every sub-stage a single-agent problem whose CCP is a *scalar* fixed point rather than a joint fixed point across regions, and it removes any coupling in the solver between regions at the same stage.

Concretely, we solve each stage in reverse region order. Region R's scalar FP is computed first for every reachable sub-state, and cached. Region $R-1$'s FP is then computed given the cached region-R continuation, and so on down to region 1. Across stages, the usual layered caches still apply: stage-4 (PE) depends only on the post-new-stage state and is reused across initial states via $\text{EV}_{\text{after pe}}[s_{an}, r]$; stages 3, 2, 1 cascade through $\text{EV}_{\text{after new}}$, $\text{EV}_{\text{after both}}$, and $\text{EV}_{\text{after old}}$, each keyed by (State, r). Because each sub-state's CCP is a function of the realized

history, the per-region CCPs reported in the results below are *marginalised* over the equilibrium play of earlier-moving regions, computed by a forward traversal of the sub-stage tree.

Because the state space now has $4R = 12$ integer coordinates, we cap $N_{\max} = 6$ on this branch (about 1.8×10^4 enumerated states) and pick initial states with total firm count ≤ 6. One visible artefact of the sequential-move convention is a small first-mover asymmetry in the baseline: with uniform $\gamma_r = 0.05$ the reported innovation CCPs are (0.3156, 0.3148, 0.3140) rather than exactly identical across regions. Region 1 commits without observing anyone, while region 3 best-responds to the realised history; the gap is well below the magnitude of every counterfactual we run and does not affect any qualitative conclusion. Two small comparative-statics exercises that probe how innovation responds to parameter or initial-state perturbations are deferred to Appendix A; they illustrate the model's mechanics but do not include welfare accounting or the estimated parameter vector.

6 Estimation on Simulated Data

As a first pass at recovering the model's structural parameters from data, we simulate a firm-level dataset from the regional model under the calibration of Table 1 and then estimate (γ, κ, ϕ), treating the demand parameters, the EVT1 scale σ, the cost $c_{n,0}$, and the state cap $N_{\max}$ as known.

6.1 Simulated dataset

The simulator (`code/scripts/simulate_data.jl`) draws 500 independent markets from the 2-period regional model. Each market gets its own initial state $s_0 \sim \text{unif}\{0, 1, 2\}^{4R}$ (rejecting empty or over-capacity draws), so the panel covers a range of local cluster sizes. For every firm in every period we record region, firm type, the two Cournot quantities, and realised profit. For period 1 we additionally record the discrete action. The structural parameters used to draw the data are deliberately *not* stored in the CSV.

6.2 Two-step estimator

Because equilibrium quantities are a deterministic function of the state and the spillover vector γ, identification splits cleanly: γ is pinned by the Cournot block and (κ, ϕ) by the discrete actions, conditional on γ. We exploit this with a two-step estimator (see `notes/estimation.md` for the full writeup).

Step 1: NLS for γ. For each market we reconstruct s_0 and s_1 by counting firms of each type and region in the corresponding period. We then solve

$$\widehat{\gamma} \;=\; \arg\min_{\gamma} \sum_{m,t,r,\,\text{slot}} \left(q^{\text{obs}}_{m,t,r,\text{slot}} - \hat{q}_{m,t,r,\text{slot}}(\gamma)\right)^2, \tag{7}$$

starting from $\gamma^0 = (0, 0, 0)$ and leaving γ *unconstrained*. Because the simulated quantities are noiseless, the minimum is machine zero (SSR $= 4.88e - 14$) and $\widehat{\gamma}$ equals the truth to all reported digits; this step doubles as a sanity check that state reconstruction is correct.

Step 2: MLE for (κ, ϕ). Holding $\gamma = \widehat{\gamma}$ fixed, we maximise the period-1 action log-likelihood over (κ, ϕ). The likelihood *replays* the sequential-move order used by the solver: within each stage, regions act in order $r = 1 \to 2 \to 3$, and each firm's contribution uses the stage CCP evaluated at the sub-state its firm actually faced (not the marginal CCP integrated over the forward traversal). Per market,

$$\begin{aligned}
\ell_m(\kappa, \phi) = &\sum_{r=1}^{R} \Big[k_r^{\mathrm{so}} \log p_{\mathrm{so}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{io}} \log p_{\mathrm{io}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{eo}} \log\big(1 - p_{\mathrm{so}} - p_{\mathrm{io}}\big) \Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{sb}} \log p_{\mathrm{sb}}(\tilde{s}_r^{\mathrm{BOTH}}) + k_r^{\mathrm{eb}} \log\big(1 - p_{\mathrm{sb}}\big) \Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{sn}} \log p_{\mathrm{sn}}(\tilde{s}_r^{\mathrm{NEW}}) + k_r^{\mathrm{en}} \log\big(1 - p_{\mathrm{sn}}\big) \Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{ep}} \log p_{\mathrm{ep}}(\tilde{s}_r^{\mathrm{PE}}) + k_r^{\mathrm{op}} \log\big(1 - p_{\mathrm{ep}}\big) \Big],
\end{aligned} \tag{8}$$

where the $k_r^{\bullet}$ are observed action counts in region r and the $\tilde{s}_r^{\bullet}$ are the sub-states produced by sequentially updating s_0 with the realised outcomes of earlier regions and stages. The total objective is $\ell(\kappa, \phi) = \sum_m \ell_m(\kappa, \phi)$, maximised by Nelder-Mead starting from $(\kappa^0, \phi^0) = (1.0, 1.0)$. The starting point is intentionally far from the calibration, so the optimiser must traverse the parameter space. Standard errors come from the BHHH estimator $V = (\sum_m g_m g_m')^{-1}$ with central finite-difference gradients.

6.3 Results

Table 4 reports the estimates against the data-generating parameters. The two-step estimator recovers $\widehat{\gamma} = (0.1500, 0.1500, 0.1500)$ exactly (Step 1), and the action MLE delivers $\widehat{\kappa} = 0.3002$ (s.e. 0.0416, truth 0.3000) and $\widehat{\phi} = 0.1777$ (s.e. 0.0407, truth 0.2000). Both κ and ϕ lie within roughly one standard error of their true values; given 500 markets and a handful of deciders per market, this is about the precision the sample supports. The period-1 action log-likelihood at the optimum is -1851.916.

7 Counterfactual: EU–US Alliance

Pooling regions 1 and 2's innovator counts lifts welfare by roughly +6.44 % in each allied region, leaves region 3 unchanged in welfare terms, and raises aggregate welfare by +4.31 % of baseline. Innovation rates rise by +5.16 % and +4.43 % in the allied regions and fall by $|-4.00|$ % in region 3, so the unallied region is harmed on the real-economy margin even though it breaks even on welfare. The rest of the section sets up the spillover specification, derives the welfare formula, describes the Monte Carlo evaluation, and walks through the mechanism.

Table 4: Two-step estimation of (γ, κ, ϕ)

Parameter	Truth	Estimate	S.E.
κ	0.3000	0.3002	0.0416
ϕ	0.2000	0.1777	0.0407
γ_1	0.1500	0.1500	—
γ_2	0.1500	0.1500	—
γ_3	0.1500	0.1500	—

Notes: Estimates from 500 simulated markets. Step 1 identifies γ from Cournot quantities; Step 2 estimates (κ, ϕ) by period-1 action MLE conditional on $\widehat{\gamma}$. Standard errors (BHHH) are reported only for the MLE step.

7.1 Spillover specification under the alliance

Under the baseline, region r's new-technology marginal cost depends only on its own innovators, equation (6). Under the alliance of $\{1, 2\}$, regions 1 and 2 share an enlarged pool:

$$c_{n,r,t}^{\text{alliance}} = c_{n,0} - \gamma_r \cdot \sum_{r' \in \text{bloc}(r)} \big(n_{b,r',t} + n_{n,r',t}\big), \qquad \text{bloc}(1) = \text{bloc}(2) = \{1, 2\},\ \text{bloc}(3) = \{3\}. \quad (9)$$

The implementation adds a single `blocs` field to the parameter struct; the default `(1,2,3)` recovers equation (6) bit-for-bit.

7.2 Welfare

The two-market linear demand of equations (1)–(2) is rationalised by the quasi-linear utility $U = A(Q_o + Q_n) - \frac{b}{2}(Q_o^2 + Q_n^2) - s\, Q_o\, Q_n$ with $b = B/M$ and $s = \rho B/M$, giving consumer surplus

$$\text{CS}(s) = \tfrac{b}{2}(Q_o^2 + Q_n^2) + s\, Q_o\, Q_n. \quad (10)$$

Because Cournot is global, CS is a single number; we allocate it equally across regions (equal-population assumption). Producer surplus is local: each firm's per-period profit is charged to its own region. Innovation and entry costs are paid by the firm's region in period 1. Two-period discounted welfare per region is

$$W_r = \tfrac{1}{R}\big[\text{CS}(s_0) + \beta\, \text{CS}(s_1)\big] + \text{PS}_r(s_0) + \beta\, \text{PS}_r(s_1) - \kappa\, k_r^{\text{innov}} - \phi\, k_r^{\text{enter}}, \quad (11)$$

which satisfies $\sum_r W_r = \text{CS}_{\text{total}} + \sum_r \text{PS}_r - \sum_r (\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}})$ by construction. The runner verifies this identity numerically and finds residuals at machine precision.

7.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario. Each market draws its own initial state s_0 from the same distribution `simulate_data.jl` uses, so the counterfactual sits on the same population the model was estimated on. The same seed (20260424) drives both

scenarios, and per-market we use `MersenneTwister(seed + k)`. Identical s_0 draws and identical EVT1 shock paths under both scenarios mean each market contributes a paired observation $(W_r^{\text{base}}, W_r^{\text{alli}})$, so the variance of the welfare difference is much smaller than the variance of either level. Headline numbers stabilise to about 0.2 percentage points by $K = 1000$.

7.4 Calibration and results

We hold the demand parameters, β, σ, ρ, c_o, $c_{n,0}$, and $N_{\max}$ at the values of Table 1, and use the structural estimates from Section 6: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 5 reports the full per-region decomposition. Innovation rises by +5.16 % in region 1 and +4.43 % in region 2, and falls by $|-4.00|$ % in region 3. Entry rates move in the same direction with smaller magnitudes.

Table 5: EU–US alliance: per-region equilibrium and welfare

	Region 1 (US)		Region 2 (EU)		Region 3 (RoW)	
Quantity	Baseline	Alliance	Baseline	Alliance	Baseline	Alliance
$P(\text{innov} \mid \text{old})$	0.3507	0.3688	0.3495	0.3650	0.3533	0.3392
$\Delta P/P_0$	+5.16%		+4.43%		-4.00%	
$P(\text{enter} \mid \text{pe})$	0.5562	0.5723	0.5801	0.5968	0.5792	0.5624
$\Delta P/P_0$	+2.91%		+2.87%		-2.91%	
PS_r	0.8500	0.9290	0.8488	0.9290	0.8800	0.8147
Costs paid	0.0905	0.0942	0.0958	0.0993	0.0958	0.0925
W_r	2.1358	2.2734	2.1293	2.2682	2.1606	2.1608
$\Delta W_r/W_{r,0}$	+6.44%		+6.52%		+0.01%	

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, uniform $\widehat{\gamma}_r = 0.1500$. Baseline blocs $(1, 2, 3)$; alliance blocs $(1, 1, 2)$. CS allocated equally across the three regions; PS_r aggregates own-region firms' profits in both periods; costs paid charges $\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}}$ to region r.

7.5 Discussion

Three points are worth highlighting.

Allied regions gain in two ways. Their pooled innovator count lowers $c_{n,1}$ and $c_{n,2}$ whenever *either* region has an innovator, raising the period-2 profit of being on new tech. This pulls more firms into innovation and entry there, which raises producer surplus, and the additional new-tech output also lowers prices globally, raising the regions' share of consumer

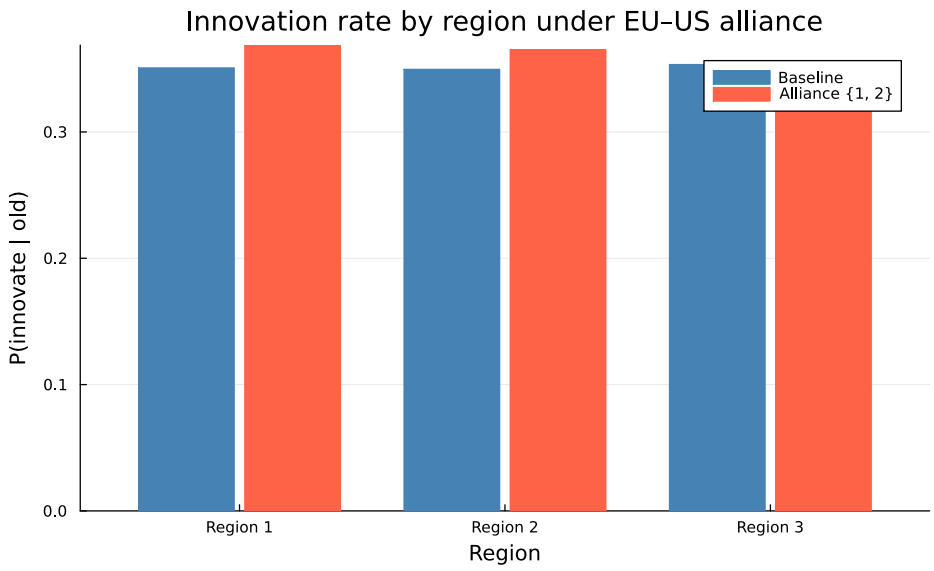

Figure 3: Innovation rate $P(\text{innov} \mid \text{old}, r)$ by region, baseline vs. alliance.

surplus. Welfare rises by +6.44 % in region 1 and +6.52 % in region 2.

Region 3 loses on innovation but breaks even on welfare. Tougher global Cournot competition from cheaper allied competitors drags region 3's $P(\text{innov})$ and $P(\text{enter})$ down by about 4 % and 3 % respectively, and its PS_3 falls by roughly 7 %. Its share of the global CS gain is almost the same magnitude in the opposite direction, so its net welfare moves by less than a basis point (+0.01 %). The unallied region is harmed in real-economy terms (less innovation, lower profits) but the consumer-side benefit of cheaper goods produced by the alliance compensates almost exactly.

The aggregate gain is positive and sizeable. Total welfare rises by +4.31 % of baseline. Roughly half of the gain accrues to each allied region, with a near-zero residual from region 3. Knowledge integration is a positive-sum game even before factoring in the aggregate consumer benefits.

The result depends on the calibration. Under an earlier $(\gamma, \sigma) = (0.05, 1.0)$ setting the same alliance moved welfare by under 1 % everywhere; tripling γ scales the spillover delivered by the merger, and halving σ doubles the logit slope $P(1-P)/\sigma$ against any payoff change, so the combined effect on equilibrium quantities is roughly 6×. The recalibrated values stay well below the cost-floor saturation that sets in at $\gamma \gtrsim 0.30$, so the model is in its responsive interior. Full details, including the K-stability diagnostic and the $\sigma \cdot \gamma_{\text{em}}$ welfare convention, are in `notes/merger_counterfactual.md`.

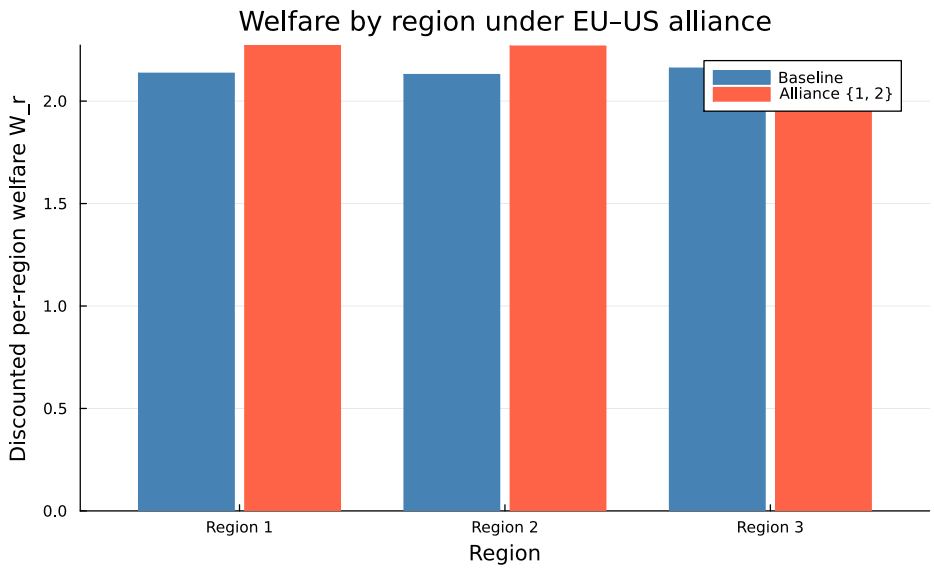

Figure 4: Per-region discounted welfare W_r baseline vs. alliance. Both allied regions gain visibly; region 3's bars are indistinguishable.

8 Next Steps

1. **Heterogeneous absorptive capacity.** The alliance result here uses uniform $\widehat{\gamma}_r$. Allowing γ_r to differ across regions, with the bloc-pooled spec of equation (9), would let one region's higher absorptive capacity amplify the alliance's effect.

2. **Fiscal cost of policy.** The current welfare accounting prices in firms' costs paid but does not net out the fiscal cost of any subsidy (e.g. the γ_1 tilt of CS-A in Appendix A). Adding government revenue/cost makes the targeted-subsidy and alliance scenarios directly comparable as policy levers.

3. $T > 2$ **horizon.** A genuine multi-period extension would force the value-function recursion to use the $\sigma \log \sum \exp$ ex-ante form at every internal period. The current 2-period setup avoids this because period 2 is terminal.

A Simple comparative-statics exercises in the regional model

These are illustrative exercises, not full counterfactuals. Both stay inside the regional model and ask how innovation responds to a small parameter or initial-state perturbation. They use a smaller baseline $\gamma = 0.05$ rather than the Table 1 default of 0.15, which lets the policy perturbation $\gamma_1 = 0.15$ act as a clean tripling of region 1's spillover and keeps every scenario well below the cost-floor saturation that sets in around $\gamma \gtrsim 0.30$. All other parameters are held at the values of Table 1. Welfare accounting and the estimated parameter vector are reserved for the proper counterfactual in Section 7.

We solve the regional model at two initial states, both with total firm count 6:

- **Symmetric s_0:** $s_0^{\text{sym}} = \big((1,1,1),(1,1,1),(0,0,0),(0,0,0)\big)$. One old incumbent and one "both" incumbent in each region, no entrants.
- **Clustered s_0:** $s_0^{\text{clust}} = \big((1,1,1),(3,0,0),(0,0,0),(0,0,0)\big)$. Same total number of firms of each type, but all three "both" incumbents are concentrated in region 1.

Starting from s_0^{sym} with uniform $\gamma_r = 0.05$ we verify a **symmetry check**: the three regions' CCPs are nearly identical, with only tiny differences (on the order of 10^{-3}) that reflect the sequential region-move structure of Section 7's solver rather than any substantive break in the model's permutation symmetry when the state and parameters are symmetric.

We then run two comparative statics against this baseline. **CS-A** subsidises innovation in region 1 by tripling its agglomeration strength to $\gamma_1 = 0.15$ while leaving $\gamma_2 = \gamma_3 = 0.05$. **CS-B** holds γ_r uniform but starts from the clustered initial state s_0^{clust}, asking whether a historical concentration of innovators is enough to lock in innovation leadership even without a policy tilt.

Table 6: Regional innovation CCPs by scenario

Scenario	$P(\text{innov}\|o, r{=}1)$	$P(\text{innov}\|o, r{=}2)$	$P(\text{innov}\|o, r{=}3)$
Baseline	0.3156	0.3148	0.3140
CS-A: subsidy r1	0.3711	0.3005	0.2999
CS-B: cluster r1	0.3228	0.2921	0.2917

Notes: Reports $P(\text{innovate} \mid \text{old}, r)$ for each region. Baseline uses s_0^{sym} with uniform $\gamma = 0.05$; CS-A sets $\gamma_1 = 0.15$ (regions 2 and 3 held at 0.05); CS-B uses the clustered initial state s_0^{clust} with uniform $\gamma = 0.05$.

Three findings stand out in Table 6 and Figure 5:

1. **Targeted subsidies work, but leak.** Raising γ_1 from 0.05 to 0.15 moves region 1's innovation probability from 0.3156 to 0.3711, a gain of roughly several percentage points. Regions 2 and 3 fall from 0.3148 to 0.3005. The spillover is small but clearly negative, driven entirely by the *global* product market: more innovation in region 1 lowers expected new-gen prices everywhere, eroding the option value of innovating in the other regions.
2. **Historical clusters alone are weak.** Starting from s_0^{clust} the incumbent cluster in region 1 barely moves innovation there (0.3228 vs. 0.3156 in the symmetric baseline): with only one remaining old firm in region 1, the region's marginal old firm is already close to saturating its own local spillover. The larger effect shows up in regions 2 and 3, whose CCPs fall to 0.2921 because of the global-market pressure from the concentrated region-1 innovators. A one-shot historical endowment is therefore a much weaker policy lever than a continuing subsidy.

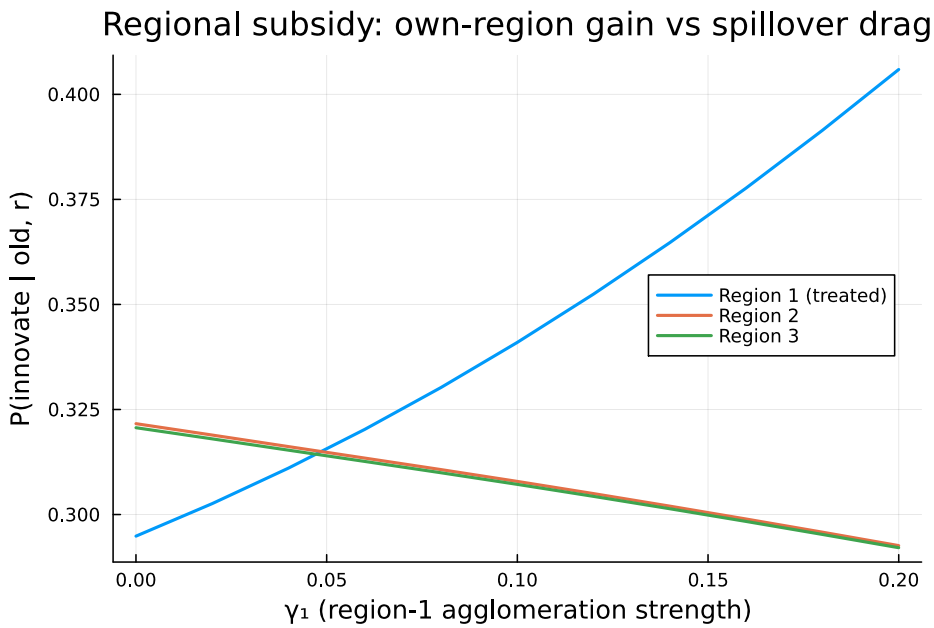

Figure 5: Regional innovation rates as γ_1 varies while $\gamma_2 = \gamma_3 = 0.05$ is held fixed. Region 1 responds positively to its own subsidy; regions 2 and 3 are slightly depressed through the global Cournot channel.

3. **The spillover wedge is still local.** Comparing CS-A and CS-B makes clear that what drives region 1's innovation is not how many innovators are in the region today but the *marginal reduction in new-tech cost* each additional innovator delivers. The policy CS-A tilts that reduction directly, which is why it dominates a pure history-based CS-B at the same total cluster size.

The two exercises illustrate how the regional model bridges Igami's replacement-effect story with the industrial-policy literature: even a modest, regionally targeted spillover subsidy reshapes the geographic distribution of innovation at the cost of a small drag on the rest of the world, with magnitudes that depend on the level of cannibalization ρ and the strength of the local externality γ_r.